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Department of Computer Science
Academic Year: 2024-25

F.Y.B.Sc.(Data Science) Sem – II
DS-152-P: Lab Course on DS-151-T (Advanced Python Programming)

Practical Assignment4

Date: 27/12/2024 and 3/1/2024

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1. **Write a Python program to read an entire text file.**

```
f = open("data.txt", "r")
content = f.read()
print(content)
f.close()
```

2. **Write a Python program to append text to a file and display the text.**

```
f = open("data.txt ", "a")
f.write("Now the file has more content!")
f.close()
f = open("abc.txt", "r")
print(f.read())
```

3. **Write a Python program to resetting the File Pointer to read again.**

```
f=open("data.txt ","r")
print(f.read())
f.seek(0)
print(f.read())
f.close()
```

4. **Write a Python program to reading a file in backward direction.**

```
f=open("data.txt ","r")
data=f.read()
print(data)
data1 = data[::-1]
print(data1)
f.close()
```

5. **Write a Python program to read first n lines of a file.**

```
n = int(input("\n\t\tEnter No. Lines To Read : "))
f = open("data.txt","r")
for i in range(n):
    print(f.readline())
```

6. **Write a Python program to read data from a Specific Position.**

```
f = open(“data.txt’, “r”)
f.seek(10)        # Set the read position to the desired location i.e. at 10th character)
content = f.read()
print("Content:", content)
f.close()
```

7. **Write a Python program to inserting Data at a Specific Position**

```
f = open(“data.txt”, “r+”)
f.seek(5)
content = f.read()
print("Content:", content)
f.seek(15)
f.write("New content")
f.close()
```

8. **Write a Python program to copy content of one file to another file.**

```
with open(“first.txt”,”r”) as f1:
   with open(“second.txt”,”a”) as f2:
      for line in f1:
         f2.write(line)
```

9. **Write a Python program to capitalize the content while writing the content into the second file.**

```
with open(“first.txt”,”r”) as f1:
   with open(“second.txt”,”a”) as f2:
      for line in f1:
         f2.write(line.upper())
```

10. **Write Python program to copy odd lines of one file to another file in Python.**

```
f1=open("first.txt","r")
f2=open("second.txt","w")
lines=f1.readlines()
for line in range(0,len(lines)):
   if(line%2!=0):
         f2.write(lines[line])
         print(lines[line])
f2.close()
f1.close()
```

11. **Write a Python program to read last n lines of a file.**

```
n=int(input("HOw no.of lines you want to read from last:"))
f = open("first.txt",'r')
lines = f.readlines()
last_lines = lines[-n:]
for line in last_lines:
   print(line)
f.close()
```

12. #Write a Python program to count the number of lines in a text file.

```
f=open("first.txt","r")
lines = len(f.readlines())
print('Total Number of lines:', lines)
f.close()
```

13. Write a Python program to compute the number of characters, words and lines in a file.

```
f=open("new.txt","r")
char_count = 0
word_count = 0
space_count = 0
line_count = 0

for line in f:
    line_count += 1
    char_count += len(line)
    words = line.split()
    word_count += len(words)
    space_count += len(words) - 1

print("Character count:", char_count)
print("Word count:", word_count)
print("Space count:", space_count)
print("Line count:", line_count)
f.close()
```

14. Write a function to read a file and print its contents. (hint -Use try-except-else to handle file not found errors.)

```
file_path = "data.txt"
try:
  with open(file_path, "r") as file:
    content = file.read()
    print(content)
except FileNotFoundError:
  print("File not found.")
```

15. Write a Python Program to perform the following:

Write Operation:

The write_to_file() function opens a file in 'write' mode ('w'), which creates the file if it doesn't exist, or truncates it (empties it) if it does exist.

The content is written into the file.

Read Operation:

The read_from_file() function opens the file in 'read' mode ('r') and reads the content.

If the file does not exist, a FileNotFoundError is caught and a message is returned

 # Function to write to a file

```
def write_to_file(filename, str):
    with open(filename, "w") as file:
        file.write(str)
    print(f"Content written to {filename}")
```

 # Function to read from a file

```
def read_from_file(filename):
    try:
        with open(filename, "r") as file:
            content = file.read()
        return content
```

```
        except FileNotFoundError:
            return f"The file {filename} does not exist."
```

 # File name

```
    filename = "example.txt“
    str = "Hello, this is a sample content for the file. \n Let's perform file operations in Python."
    write_to_file(filename, str)
    file_content = read_from_file(filename)
    print("\nContent read from file:")
    print(str)
```